PREVIEW

CAMBRIDGE O LEVEL

COMPUTER SCIENCE 2210

PAPER 1 · COMPUTER SYSTEMS



TOPICAL PAST PAPERS

2016 – 2026 · WITH MARK SCHEMES

- 8 chapters
- 394 questions
- Full writing space
- Mark schemes after all questions

Preview

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CONTENTS

No.	Chapter	Questions	Q. page	MS page
1	Data representation	89	3	388
2	Data transmission	61	78	412
3	Hardware	134	129	433
4	Software	45	219	475
5	The internet and its uses	78	248	489
6	Automated and emerging technologies	33	316	518
10	Boolean logic	27	340	529
E	Extra: topics not in the 2026–2028 syllabus	18	376	544

485 entries from 394 questions · 41 Paper 1 past papers (May/June and Oct/Nov, 2016–2026)
Chapters 7–9 (algorithms, programming and databases) are Paper 2 topics, so they are not in this Paper 1 book.

MY PROGRESS

Shade one box for every 10% of a chapter you finish.

	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
1 Data representation										
2 Data transmission										
3 Hardware										
4 Software										
5 The internet and its uses										
6 Automated and emerging technologies										
10 Boolean logic										
E Extra topics										

Preview

CHAPTER

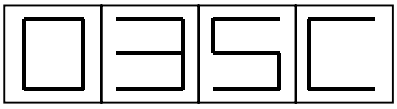
01 Data representation

89 questions · 2016–2026 · mark schemes at the back

1010
0110

Q1 2016 · May/June · Paper 11 · Q7 done

7 Each seat on a flight is uniquely identified on an LCD above the seat. For example, seat 035C is shown as:



The first three characters are digits that represent the row.
The fourth character is the seat position in that row. This is a single letter, A to F, that is stored as a hexadecimal value.

Each of the four display characters can be stored in a 4-bit register. For example, 0 and C would be represented as:

	8	4	2	1
0:	0	0	0	0
C:	1	1	0	0

(a) Show how the 4-bit registers would store the remaining two characters, 3 and 5.

3 ☐☐☐☐

5 ☐☐☐☐

[2]

(b) Identify which seat is stored in the following 4-bit registers.

0	0	0	1	→ .....
1	0	0	1	→ .....
0	1	0	0	→ .....
1	1	1	0	→ .....

[2]

Preview

Q2 2016 · May/June · Paper 11 · Q12 **Part (b)** *+ (a) for context* done ☐

12 **(a)** Name the following type of barcode:



.....[1]

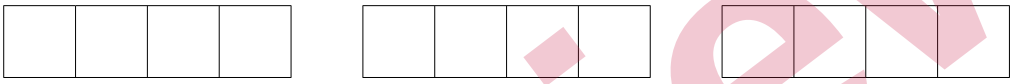
(b) The barcode in **part (a)** contains the denary value 2 6 4 0

Convert this value to hexadecimal.

.....

.....

Write the value as a 12-bit binary number.



[4]

Part (c) of this question is in Chapter 3

Q3 2016 · May/June · Paper 12 · Q3 done ☐

3 **(a)** Convert the following hexadecimal number into 12-bit binary:

4 A F



[3]

(b) The 2016 Olympic Games will be held in Rio de Janeiro. A timer that counts down to the opening of the Games is shown on a microprocessor-controlled display.

The number of hours, minutes and seconds until the Games open are held in three 8-bit registers.

The present register values are:

0	1	1	0	1	0	0	1	105 hours
0	0	1	0	0	0	0	0	32 minutes
0	0	0	1	0	1	0	0	20 seconds

The timer will count **down** in seconds.

Preview

Q3 2016 · May/June · Paper 12 · Q3 *continued* done ☐

(i) Show the values in each 8-bit register **30 seconds** after the time shown above:

								hours
								minutes
								seconds

[3]

(ii) Write the hexadecimal value of the **minutes** register from **part (b)(i)**.

.....[1]

Q4 2016 · May/June · Paper 12 · Q4 **Part (a)** done ☐

4 Nigel wants to send a large text file electronically to Mashuda.

(a) Describe how the size of the text file can be reduced.

.....

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.....

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.....

.....

.....[3]

Part (b) of this question is in Chapter 2

Preview